MIT 18.06

Linear Algebra

Problem Set 1

#1 Section 1.2 - Q 23

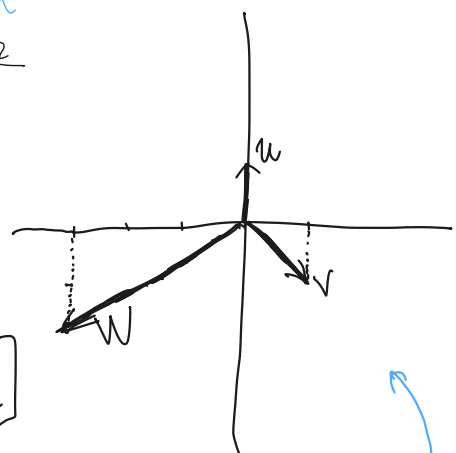
$\cos\alpha = \frac{v_1}{\|v\|}$ $\sin\alpha = \frac{v_2}{\|v\|}$

$\cos\beta = \frac{w_1}{\|w\|}$ $\sin\beta = \frac{w_2}{\|w\|}$ $\theta = \beta - \alpha$

$\cos\theta = \cos(\beta-\alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha$

$= \frac{w_1}{\|w\|}\frac{v_1}{\|v\|} + \frac{w_2}{\|w\|}\frac{v_2}{\|v\|} = \frac{w_1v_1 + w_2v_2}{\|w\|\|v\|}$

$= \frac{w\cdot v}{\|w\|\|v\|} = \frac{v\cdot w}{\|v\|\|w\|}$



#2 Section 1.2 - Q 28

$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$, $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$

Let's set $u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then for $u\cdot v < 0$, $u_1v_1 + u_2v_2 < 0$

$0\cdot v_1 + 1\cdot v_2 < 0$, so $v_2 < 0$, for $u\cdot w < 0$, using a similar process, we get $w_2 < 0$, for $v\cdot w < 0$, since $v_2\cdot w_2 > 0$, we have to make $v_1\cdot w_1 < 0$ and also $|v_1\cdot w_1| < v_2\cdot w_2$, say $v_2 = -1$, $w_2 = -2$, $v_2w_2 = 2$

we can put $v_1 = 1$, $w_1 = -3$ to satisfy the conditions,

so $u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $w = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$

$uv = 0\cdot 1 + 1\cdot(-1) = -1$, $vw = 1\cdot(-3) + (-1)\cdot(-2) = -1$, $uw = 0\cdot(-3) + 1\cdot(-2) = -2$

a geometric image is shown above.

as long as the angle between them is obtuse.

#3 Section 1.3 – Q4

$$w_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad w_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad w_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

$$w_1 - 2 \cdot w_2 = \begin{bmatrix} 1-2\cdot 4 \\ 2-2\cdot 5 \\ 3-2\cdot 6 \end{bmatrix} = \begin{bmatrix} -7 \\ -8 \\ -9 \end{bmatrix} = -w_3, \text{ so } w_1 + w_3 = 2w_2,$$

These vectors are dependent. The three vectors lie in a plane. The matrix W with those columns is not invertible

#4 Section 1.3-Q13

$$Gx = b \qquad \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 \\ -x_1 + x_3 \\ -x_2 + x_4 \\ -x_3 + x_5 \\ -x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

we notice $b_1 + b_5 = -b_3$ so $b_1 + b_3 + b_5 = 0$

#5 Section 2.1 – Q29

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

$$u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad u_1 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

$$u_2 = Au_1 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.8\cdot 0.8 + 0.3\cdot 0.2 \\ 0.2\cdot 0.8 + 0.7\cdot 0.2 \end{bmatrix} = \begin{bmatrix} 0.64 + 0.06 \\ 0.16 + 0.14 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

$$u_3 = Au_2 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix} = \begin{bmatrix} 0.8\cdot 0.7 + 0.3\cdot 0.3 \\ 0.2\cdot 0.7 + 0.7\cdot 0.3 \end{bmatrix} = \begin{bmatrix} 0.56 + 0.09 \\ 0.14 + 0.21 \end{bmatrix} = \begin{bmatrix} 0.65 \\ 0.35 \end{bmatrix}$$

Property: sum to 1

#6 Section 2.1 – Q30

See the file ps1-6.ipynb
in problem_code folder

Based on the visualization, I guess $S = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$

$$AS = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.8 \cdot 0.6 + 0.3 \cdot 0.4 \\ 0.2 \cdot 0.6 + 0.7 \cdot 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.48 + 0.12 \\ 0.12 + 0.28 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = S, \text{ so } AS = S$$

#7 Section 2.2 – Q20

- linear combination
- a 3rd equation: $2x - y = 2$ or $3y + 2z = -2$, etc.

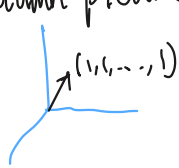
#8 Section 2.2 – Q32

(a) zero (b) zero

(c) a 100 by 100 matrix full of 1's

(d) – row picture
100 copies of $x_1 + x_2 + \cdots + x_{100} = 0$ hyperplane passing through the origin

– column picture



$$\sqrt{\underbrace{1^2 + 1^2 + \cdots + 1^2}_{100}} = 10$$

100 vectors $(\underbrace{1, 1, \ldots, 1}_{100})$ all have length 10.

#9 Section 2.3 - Q22

(a) $\sum a_{3j}x_j = a_{31}x_1 + \cdots + a_{3n}x_n$

(b) $a_{21} - a_{11}$ (c) $a_{21} - 2 \cdot a_{11}$

(d) $\sum a_{1j}x_j = a_{11}x_1 + \cdots + a_{1n}x_n$

#10 Section 2.3 - Q29

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

#11 Section 2.4 - Q32

$$Ax = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

#12 Section 2.4 - Q36

(a) $(AB)C \to mnp + mpq = 2 \cdot 4 \cdot 7 + 2 \cdot 7 \cdot 10 = 56 + 140 = 196$

$A(BC) \to mnq + npq = 2 \cdot 4 \cdot 10 + 4 \cdot 7 \cdot 10 = 80 + 280 = 360$

I prefer $(AB)C$

(b) $u^T \to 1 \times n$, $v \to n \times 1$, $w^T \to 1 \times n \Rightarrow m = 1$, $n = n$, $p = 1$, $q = n$

$(u^T v) w^T \to mnp + mpq = n + n = 2n$

$u^T (v w^T) \to mnq + npq = n^2 + n^2 = 2n^2$

I choose $(u^T v) w^T$

(c) $(AB)C \to mnp + mpq \xrightarrow{\text{divide by } mnpq} q^{-1} + n^{-1}$

$A(BC) \to mnq + npq \xrightarrow{\text{divide by } mnpq} p^{-1} + m^{-1}$

When $n^{-1} + q^{-1} < m^{-1} + p^{-1}$, $(AB)C$ is faster

#13 Section 2.5 - Q7

(a) $A = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix}$, for $Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $A_1 x = 1$, $A_2 x = 0$, $A_3 x = 0$,

but $A_3 x = (A_1 + A_2) x$, while $A_1 x + A_2 x = 1 \neq 0$, contradiction

(b) As long as $b_3 = b_1 + b_2$

(c) It will become zero